

Exact Verification of the Kasami Cyclic-Additive Identity at $n = 14$

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Abstract

Let $K = \mathbb{F}_{2^{14}}$. For $k \in \{3, 5\}$, put $F_k(t) = t^{2^{2k}-2^k+1}$ and $\Delta_k = \{F_k(t) + F_k(t+1) + 1 : t \in K\}$. We give an exhaustive exact-arithmetic verification that $|\Delta_k| = 8192$ and that, for every $\rho \in K \setminus \{0, 1\}$,

$$\#\{(x, y, z) \in \Delta_k^3 : x + \rho y + (1 + \rho)z = 0\} = 2^{25}.$$

The field is realized explicitly as $\mathbb{F}_2[\alpha]/(\alpha^{14} + \alpha^5 + 1)$. A complete certificate contains both 16384-entry additive-character spectra and every one of the 32764 normalized counts. A separately structured exact checker reconstructs all spectrum and correlation entries. The result resolves only the finite $n = 14$ layer; it gives no proof for arbitrary n .

1 Context and scope

Nagy and Vajda formulate the following cyclic-additive conjecture for the Kasami function on $\mathbb{F}_{2^n}$: if $\gcd(k, n) = 1$, then the corresponding set Δ has size 2^{n-1} , and every pair of distinct nonzero coefficients gives exactly 2^{2n-3} ordered triples in Δ^3 satisfying the associated linear relation [1]. They prove the residue classes $k \equiv \pm 1, \pm 2 \pmod{n}$, verify all admissible pairs for $n \leq 13$, and establish a Frobenius transfer between k and $n - k$ [1].

For $n = 14$, the admissible residues are 1, 3, 5, 9, 11, 13. The proved classes cover 1, 13, while Frobenius transfer pairs 3 with 11 and 5 with 9. Thus verification for $k = 3, 5$ completes this next finite layer relative to the coverage in [1]. Nothing below addresses the remaining general conjecture.

Exact finite-field boundary results require unusually explicit scope and reproducibility statements. As a methodological comparison only, Zeng, Liu, and Ratnavelu likewise separate a finite-field boundary theorem from an exact-arithmetic companion verifier in an unrelated permanent-rank problem [2]. Their paper contains no Kasami, Walsh-spectrum, or cyclic-additive result, and no technical statement from it is used here.

2 The finite assertion and the field model

Let

$$p(X) = X^{14} + X^5 + 1, \quad K = \mathbb{F}_2[\alpha]/(p(\alpha)).$$

We use the polynomial basis $(1, \alpha, \dots, \alpha^{13})$. The integer $\sum_{i=0}^{13} b_i 2^i$ encodes $\sum_{i=0}^{13} b_i \alpha^i$, so field addition is bitwise exclusive-or. In particular, the modulus has bit encoding 0x4021.

Lemma 1. *The polynomial $p(X) = X^{14} + X^5 + 1$ is irreducible over $\mathbb{F}_2$.*

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Proof. Exact squaring and polynomial reduction give

$$X^{2^2} \equiv 0\mathbf{x}0010, \quad X^{2^7} \equiv 0\mathbf{x}08\mathbf{c}5, \quad X^{2^{14}} \equiv 0\mathbf{x}0002 = X \pmod{p}.$$

Moreover,

$$\gcd(X^{2^2} - X, p) = \gcd(X^{2^7} - X, p) = 1.$$

The prime divisors of 14 are 2 and 7, so these are precisely Rabin's irreducibility conditions for a polynomial of degree 14. The certificate producer performs these operations in $\mathbb{F}_2[X]$. Independently of this route, the checker divides p by every monic constant-one polynomial of degrees 1 through 7, a total of $1 + 2 + \dots + 64 = 127$ candidates, and finds no divisor. $\square$

For $k \in \{3, 5\}$, define

$$d_k = 2^{2k} - 2^k + 1, \quad \delta_k(t) = t^{d_k} + (t+1)^{d_k} + 1, \quad \Delta_k = \delta_k(K).$$

Thus $d_3 = 57$ and $d_5 = 993$. For $\rho \in K \setminus \{0, 1\}$, write

$$N_k(\rho) = \#\{(x, y, z) \in \Delta_k^3 : x + \rho y + (1 + \rho)z = 0\}.$$

Theorem 2. *For both $k = 3$ and $k = 5$, one has $|\Delta_k| = 8192$, and*

$$N_k(\rho) = 33554432 = 2^{25} \quad \text{for every } \rho \in K \setminus \{0, 1\}.$$

Consequently the normalized $n = 14$ assertion holds for all $2(2^{14} - 2) = 32764$ pairs (k, ρ) in its stated domain.

Because finite fields of the same order are isomorphic and the assertion uses only field operations, verification in the model of Lemma 1 suffices. The explicit model is retained to make every certificate entry reproducible.

3 Character reduction

Let $q = 2^{14}$, let $\text{Tr}(u) = \sum_{i=0}^{13} u^{2^i} \in \mathbb{F}_2$ be the absolute trace, and define

$$S_k(a) = \sum_{x \in \Delta_k} (-1)^{\text{Tr}(ax)} \quad (a \in K).$$

Lemma 3. *For every $\rho \in K \setminus \{0, 1\}$,*

$$N_k(\rho) = \frac{1}{q} \sum_{a \in K} S_k(a) S_k(a\rho) S_k(a(1 + \rho)).$$

Proof. For $u \in K$, additive-character orthogonality gives

$$\mathbf{1}_{u=0} = \frac{1}{q} \sum_{a \in K} (-1)^{\text{Tr}(au)}.$$

Insert $u = x + \rho y + (1 + \rho)z$, sum over $(x, y, z) \in \Delta_k^3$, and factor the three finite sums. This gives the displayed identity exactly, with no approximation or limiting step. $\square$

Since $S_k(0) = |\Delta_k|$, Lemma 3 separates the desired count into the zero-character term and a correlation:

$$qN_k(\rho) = |\Delta_k|^3 + C_k(\rho), \quad C_k(\rho) = \sum_{a \in K^*} S_k(a) S_k(a\rho) S_k(a(1 + \rho)). \quad (1)$$

4 Exhaustive exact certificate

The certificate producer is the self-contained C11 program `verify_n14.c`. After the field check of Lemma 1, it performs the following finite operations.

Image sets. For each exponent, the program evaluates $\delta_k(t)$ at all 16384 encoded values of t , using shift-and-reduce polynomial multiplication and binary exponentiation. It obtains exactly 8192 image values in each case. It also checks all fiber multiplicities: every image value occurs exactly twice, and every nonimage value occurs zero times.

Spectra. For each encoded a , the program forms the dual mask whose i -th bit is $\text{Tr}(a\alpha^i)$. The resulting map from field elements to bit masks is checked to be bijective. A Walsh–Hadamard transform of the membership vector, reindexed by this dual mask, therefore gives the actual character sum $S_k(a)$, rather than a coordinate dot-product surrogate. Parseval’s identity and nine direct character sums per exponent are checked internally. All 32768 spectrum entries are written to the tab-separated certificate.

All parameters. The program enumerates the 16382 admissible encodings of ρ for each k , evaluates $C_k(\rho)$ from the saved spectrum, checks divisibility in (1), and writes the exact correlation, full character sum, and quotient. The complete outcome is

k	d_k	$ \Delta_k $	$\#\{S_k(a)\}$	$\#\{\rho\}$	$\min N_k = \max N_k$
3	57	8192	16384	16382	33554432
5	993	8192	16384	16382	33554432

and every one of the 32764 values $C_k(\rho)$ is exactly zero. Consequently

$$qN_k(\rho) = 8192^3 = 2^{39}, \quad N_k(\rho) = 2^{39-14} = 2^{25},$$

which proves Theorem 2 once the finite computation is audited.

All decisive quantities are integers. Since $|S_k(a)| \leq 8192$, even the naive absolute bound for an intermediate correlation is less than 2^{53} , well within a signed 64-bit integer. Floating-point arithmetic is not used to build the field, the sets, the spectra, the correlations, or the certificate.

Separately structured reconstruction

The C11 program `audit_n14.c` checks the certificate using different arithmetic and transform routes. It uses carryless polynomial products followed by long division, rather than the producer’s shift-and-reduce and log-table route. It reconstructs both image sets, directly recomputes every one of the 32768 character sums, parses every correlation row, and recomputes all 32764 correlations through polynomial-basis linear multiplication maps. All comparisons pass.

As a further check on the character reduction, direct pair enumeration uses the fact that, for fixed (y, z) , the equation has the unique candidate $x = \rho y + (1 + \rho)z$. The producer directly checks $\rho = 0x0002, 0x2000$, and the separate checker uses the disjoint values $\rho = 0x1555, 0x3ffe$, in both exponent cases. Every direct count is 33554432. These spot checks supplement, rather than replace, the exhaustive correlation coverage.

5 Reproduction, availability, and limitations

The source package accompanying this note contains the producer, checker, and full certificate:

```
evidence/verify_n14.c
evidence/audit_n14.c
evidence/n14_certificate.tsv
```

From the package root, a clean reconstruction is:

```
cc -std=c11 -O3 -Wall -Wextra -Wpedantic \
    evidence/verify_n14.c -o verify_n14
./verify_n14 n14_certificate.rebuilt.tsv
cmp n14_certificate.rebuilt.tsv evidence/n14_certificate.tsv

cc -std=c11 -O3 -Wall -Wextra -Wpedantic \
    evidence/audit_n14.c -o audit_n14
./audit_n14 n14_certificate.rebuilt.tsv
```

The recorded producer and checker builds were warning-free, and sanitizer builds reported no address or undefined-behavior violations. Such software checks support implementation reliability but do not replace the exact character argument or the exhaustive row coverage.

The result is a finite verification at $n = 14$. It does not establish the conjecture for any untested n , identify a uniform mechanism, or by itself establish priority. The closest technical comparison used here is the $n \leq 13$ verification and structural reduction of Nagy and Vajda [1].

Computational and AI assistance. The proof depends on the exact C11 computation and machine-readable certificate described above. AI assistance was used in developing and reviewing code and in preparing the exposition. The mathematical scope, character reduction, field construction, complete row domains, and decisive artifacts were subsequently checked in a separate mathematical audit. This is a reproducibility and review disclosure, not a claim of formal proof or independent peer review.

References

- [1] Gábor P. Nagy and Attila Vajda. On a conjecture on the Kasami APN function: Reductions, structure theorems, a proof for $k \bmod n \in \{1, 2, n - 2, n - 1\}$, and exhaustive verification for $n \leq 13$, 2026. arXiv:2608.18584v2. URL: <https://arxiv.org/abs/2608.18584>, arXiv:2608.18584, doi:10.48550/arXiv.2608.18584.
- [2] Zijian Zeng, Houde Liu, and Kuru Ratnavelu. Closing the permanent-rank boundary for the Alon–Jaeger–Tarsi conjecture. *Preprints*, 2026. Version 1, posted 18 August 2026; not peer reviewed. doi:10.20944/preprints202608.1194.v1.